

SMARTDRIVE

AI-Powered Car Rental & Mobility Management System

PROJECT

Database Systems Engineering & Distributed Backend Development

Section 11 | Team 09 | Y25–2026–2027 Trimester 4

TEAM MEMBERS

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Abstract

What SmartDrive is designed to solve

CORE IDEA

Simplifies and automates car rental through independent user search, reservation and rental management.

Provides secure authentication, location/date-based search, booking management, simulated payments, booking history and notifications.

Uses AI-based recommendations based on user preferences and rental requirements.

SEARCH

Vehicle discovery

BOOK

Rental management

PAY

Simulated payment

AI

Personalized match

Problem Statement

Pain points in conventional car-rental workflows

Manual booking

Calls and visits can make booking time-consuming.

Limited visibility

Vehicle status may be difficult to check.

Staff dependency

Limited self-service increases operational dependency.

Data integration

Isolated systems can create integration difficulties.

Poor UX

Limited digital convenience affects the rental journey.

Selection difficulty

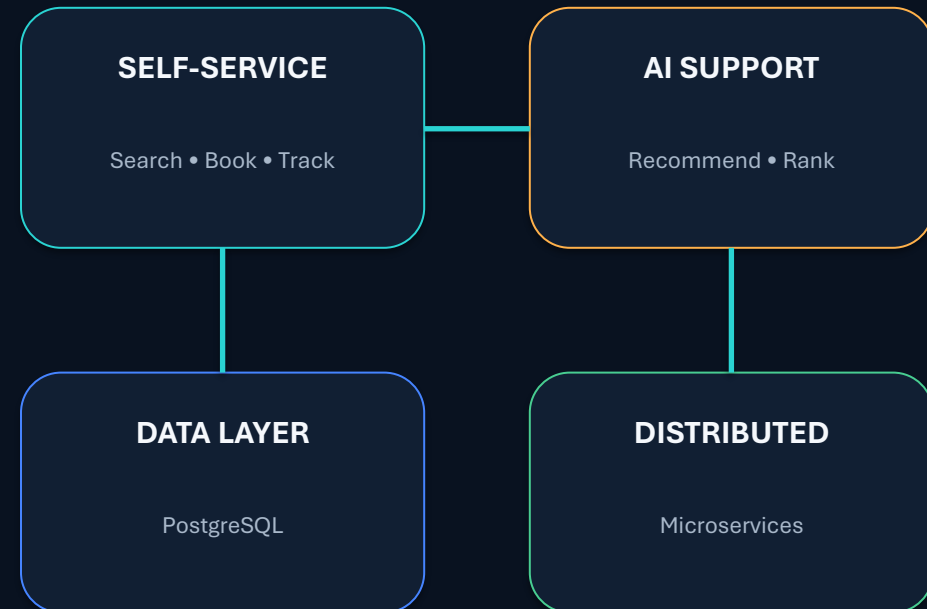
Budget, passengers, duration and preferences complicate choice.

Motivation

Why the project combines self-service, AI recommendations and distributed services

MOTIVATING NEEDS

- Reduce dependency on manual rental interactions.
- Make availability and booking easier to access digitally.
- Support personalized vehicle selection.
- Create a scalable backend through service isolation.
- Demonstrate database systems engineering in a mobility use case.



Objectives

Target capabilities defined for SmartDrive

PROJECT OBJECTIVES

Provide a user-friendly web/mobile rental interface.

Implement secure JWT-based authentication.

Enable location/date-based vehicle search and real-time availability.

Support booking management, payment processing and booking history.

Add notifications and rental tracking.

Recommend suitable cars from user requirements.

Use scalable microservices and PostgreSQL data storage.

Literature Survey

Theme-based survey framing; paper-specific evidence is not included in the supplied PPT

Digital rental platforms

Self-service search, reservations and rental tracking.

Recommendation systems

Preference-based matching for vehicle selection.

Distributed backends

Service isolation across core rental functions.

Database-backed systems

PostgreSQL for persistent platform data.

Secure web APIs

REST APIs and JWT authentication.

Research Gap

Gap identified directly from the documented problem analysis

OBSERVED NEED

Manual and staff-dependent workflows.

Limited unified availability visibility.

Difficulty matching vehicles to multiple requirements.

Integration challenges across separate operational functions.

PROPOSED RESPONSE

Unified digital search and booking.

Service isolation for authentication, vehicles, bookings, payments and notifications.

Recommendation module matching requirements with vehicle attributes.

PostgreSQL-backed persistent data layer.

Proposed Methodology

End-to-end flow from requirements to rental management

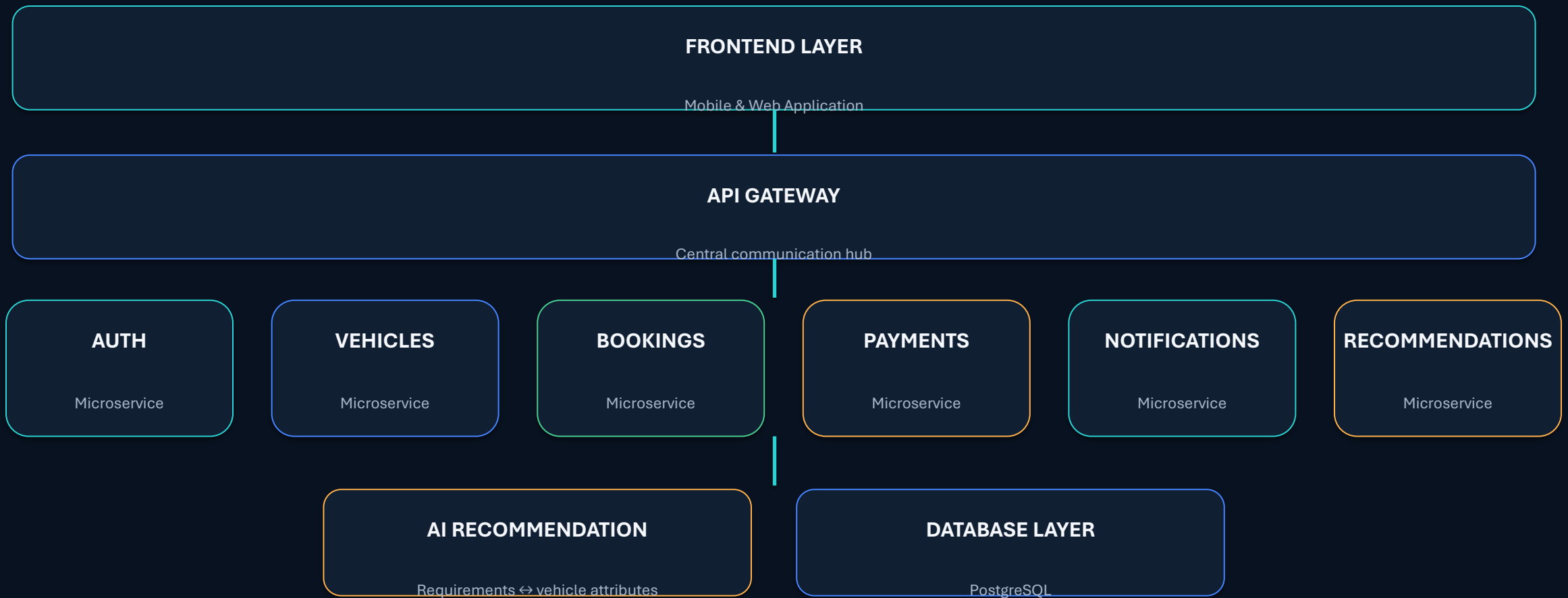


AI RECOMMENDATION PATH

Collect requirements → compare available vehicles → calculate suitability → rank matches → display recommendations.

System Architecture

Layered, distributed design documented for SmartDrive



Technologies Used

Technology stack specified in the project documentation

Frontend

HTML • CSS • JavaScript

Backend

Node.js • Express.js • REST APIs

Database

PostgreSQL

Security

JWT Authentication

AI / Recommendation

Preference-based vehicle recommendation logic

DevOps

Docker Containers

Tools

VS Code • Postman

Dataset / Input Details

Inputs used by the documented recommendation and rental workflow

USER / RECOMMENDATION INPUTS

- Budget
- Car type
- Number of passengers
- Rental duration
- User preferences
- Previous bookings

SYSTEM-SIDE DATA

- Vehicle attributes
- Vehicle availability
- Location and rental date
- Booking information
- Payment information
- User history / preferences

Implementation – Module 1

Authentication & user self-service

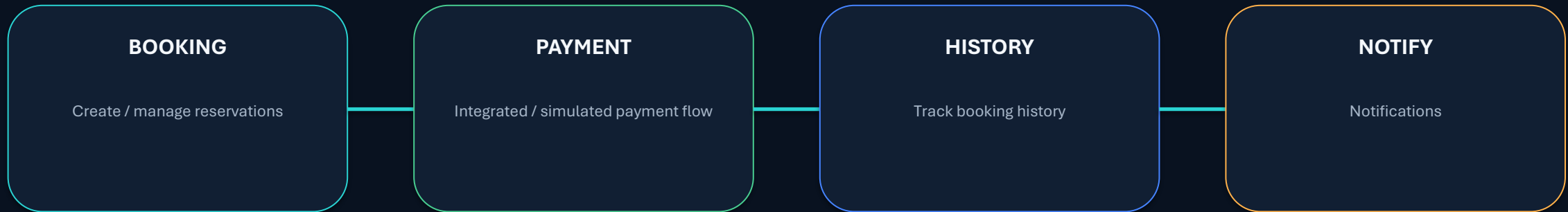


MODULE RESPONSIBILITY

Documented scope: secure authentication, user-friendly interface, vehicle search and availability.

Implementation – Module 2

Booking, payment and rental management

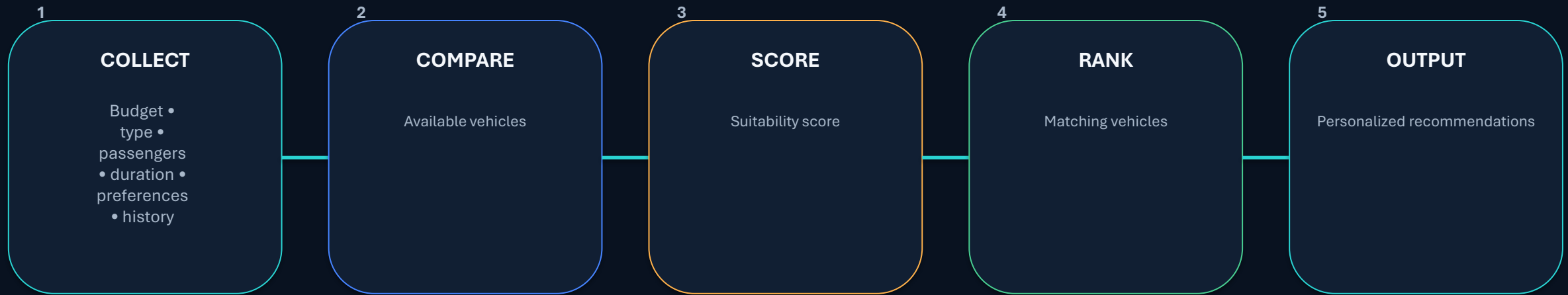


DOCUMENTATION STATUS

The source documents these as platform capabilities; detailed implementation code or transaction metrics are not included.

Implementation – Module 3

AI-based car recommendation module



OUTPUT

Personalized car recommendations intended to help users choose a suitable vehicle faster.

Experimental Setup

Evaluation plan aligned with documented system capabilities

FUNCTIONAL TESTS

- Test authentication and access flow.
- Verify location/date vehicle search.
- Validate booking creation and history.
- Exercise payment and notification flows.
- Run recommendation inputs across different requirements.

MEASUREMENT PLAN

- Measure recommendation suitability / ranking quality when evaluation data is available.
- Measure API response time and service behavior.
- Check database operations and consistency.
- Check containerized service deployment.
- Record screenshots and demo evidence.

Results

Documented outcomes and current evidence

SEARCH

Self-service discovery

BOOKING

Rental workflow

AI

Recommendation logic

API

Distributed backend

IMPORTANT INTERPRETATION

These are documented system capabilities, not measured benchmark results. The source PPT provides no accuracy, latency, throughput or user-study values.

Performance Evaluation

Metrics to report after implementation testing

Recommendation

Suitability / ranking quality

Backend

API response time

Database

Query / transaction behavior

Scalability

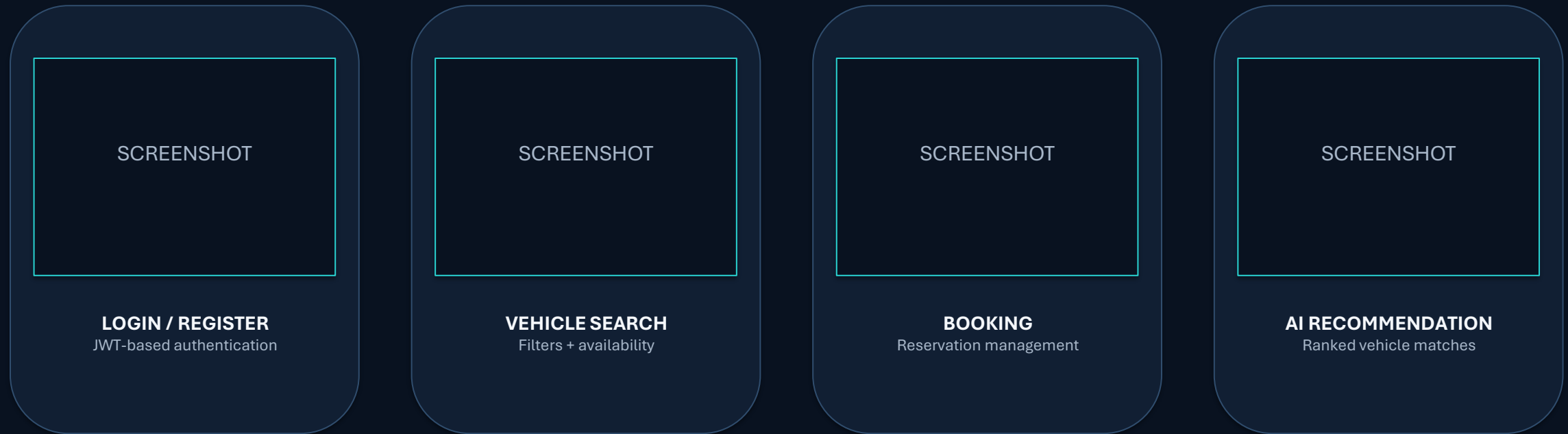
Service-level deployment behavior

Usability

Ease of search and booking

Screenshots / Working Demo

Suggested evidence frames for the final demonstration



Comparison with Existing System

Requirement-to-solution comparison based on the problem analysis

Capability	Documented pain point	SmartDrive approach
Booking	Manual calls / visits	Digital self-service booking
Availability	Limited visibility	Location/date-based availability
Staff dependency	Higher involvement	User self-service
Selection	Difficult multi-factor choice	AI-based recommendation
Integration	Isolated systems	API gateway + microservices
Data	Distributed operational data	PostgreSQL data layer

Challenges & Solutions

Design responses to the documented problem areas

Manual workflows

Self-service web/mobile interface

Vehicle selection

AI preference-based recommendation

Service integration

API gateway + REST APIs

Scalability

Microservices + Docker

Secure access

JWT authentication

Persistent data

PostgreSQL

Current Progress

Status visible from the supplied SmartDrive project documentation

System concept	Defined
Architecture	Defined
Feature set	Defined
AI recommendation flow	Defined
Technology stack	Defined
Measured results	Not provided
Demo screenshots	Not provided

Future Work

Natural extensions of the documented SmartDrive scope

SYSTEM ENHANCEMENTS

Expand recommendation logic with richer user and vehicle signals.

Add stronger evaluation with real usage data and measurable benchmarks.

Enhance notification and rental tracking workflows.

Strengthen service monitoring and operational observability.

VALIDATION & DOCUMENTATION

Add detailed performance and scalability testing.

Capture working-demo screenshots and user-flow evidence.

Document API contracts and database design in greater detail.

Validate recommendation quality with defined evaluation metrics.

Conclusion

SmartDrive brings the documented capabilities together in one distributed rental platform

KEY TAKEAWAYS

Combines car-rental self-service with distributed backend services.

Covers authentication, availability, booking, payments, notifications and booking history.

AI recommendation matches user requirements with vehicle attributes and ranks suitable choices.

Demonstrates database systems engineering, REST API design, microservices and intelligent decision support.

REFERENCES

Supplied SmartDrive project PPT / abstract

Node.js & Express.js documentation

PostgreSQL documentation

JWT authentication documentation

Docker documentation

THANK YOU • QUESTIONS & DISCUSSION